Task 3: API Integration Project

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Domain: Frontend Web Development

Objective

The objective of this project is to integrate a public REST API into a web application and display the fetched data dynamically on the webpage.

In this task, a Weather Application was developed using HTML, CSS, and JavaScript. The application fetches real-time weather information from a public API and displays it to the user.

Tools & Technologies Used

The following technologies were used to build this project:

- HTML5 – for structuring the webpage
- CSS3 – for styling the user interface
- JavaScript – for API integration and DOM manipulation
- Fetch API – for retrieving data from the API
- GitHub – for version control and code hosting
- GitHub Pages – for deploying the live project

API Used

This project uses the **OpenWeatherMap API**.

API Website:

https://openweathermap.org/api

The API provides weather data including:

- Temperature
- Weather conditions
- Humidity
- Wind speed

The application sends a request to the API and retrieves weather data in **JSON format**.

Features of the Application

The weather application includes the following features:

- Fetch weather data using Fetch API
- Display temperature and weather condition
- Search weather by city name
- Loading state while fetching data
- Error handling if city is not found
- Simple and responsive interface

Project Structure

```
Task3-Weather-API-App
│
├── index.html
├── style.css
├── script.js
└── README.md
```

Code Snippet

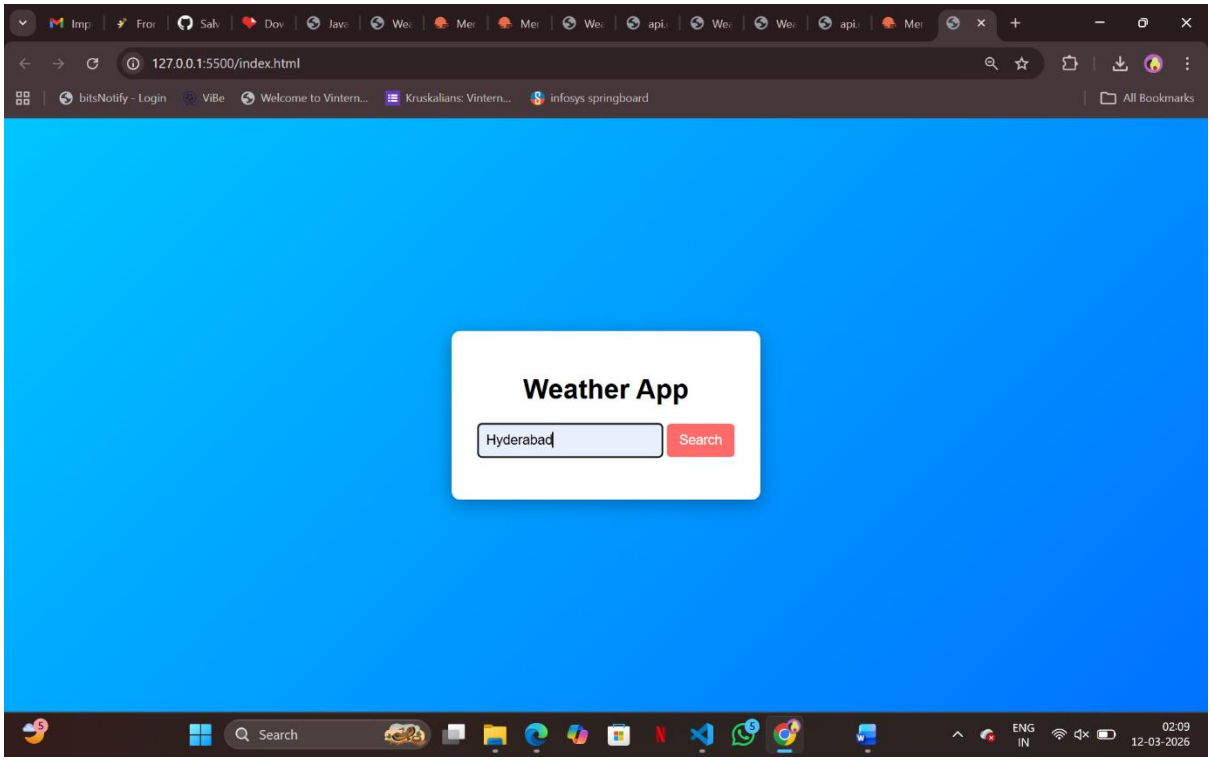
JavaScript API Fetch Example:

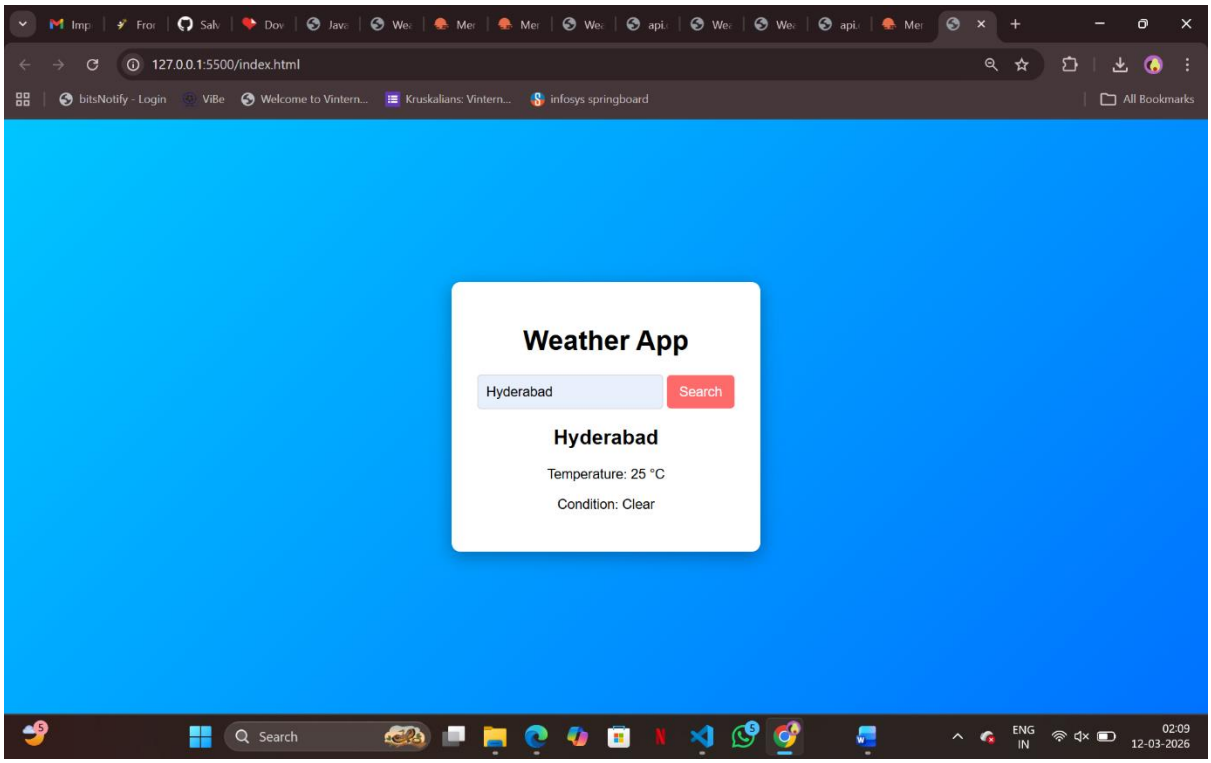
```
async function getWeather(){

let city = document.getElementById("cityInput").value;

let response = await fetch(`https://wttr.in/${city}?format=j1`);

let data = await response.json();

document.getElementById("cityName").innerText = city;

document.getElementById("temperature").innerText =

"Temperature: " + data.current_condition[0].temp_C + " °C";

document.getElementById("condition").innerText =

"Condition: " + data.current_condition[0].weatherDesc[0].value;

}
```

Screenshots

It includes:

- Weather search interface
- Weather result display
- Error message when city is invalid





GitHub Repository

Source code available at: https://github.com/Salva401/Internspark-frontend-development-tasks/tree/main/Task3-Weather-API-App

Live Website

The project is hosted using GitHub Pages: https://salva401.github.io/Internspark-frontend-development-tasks/Task3-Weather-API-App/

Conclusion

This project demonstrates how to integrate a public REST API into a web application. The weather app retrieves real-time data from an external API and dynamically displays it on the webpage using JavaScript.

The project also demonstrates understanding of **API requests, JSON data handling, and DOM manipulation**.

Author

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